

Lab Assignment 1 (Deadline 12th Asadh 2083)

1. Write a program to find prefixes, suffixes, and substring from the string. Your program should ask the user to input the string then display the prefixes, suffixes, and substring of the input string.
2. Write a program to implement a DFA such that it should accepts all the strings which has 2nd symbol as 'a' and 4th symbol as 'b' over alphabet $\Sigma=\{a, b\}$. Your program should ask the user to input the string and check whether the user input is accepted or rejected by the DFA. Also show the processing of the string in the output section.
3. Write a program to implement DFA such that it accepts the string that does not contain substring as 00, over alphabet $\Sigma=\{0,1\}$. Your program should ask the user to input the string and check whether the user input is accepted or rejected by the DFA. Also show the processing of the string in the output section.
4. Write a program to implement DFA such that it accepts the multiline comments of a C-program. Your program should ask the user to input the comment and check whether the comment is accepted or rejected by that DFA. Also show the processing of the string in the output section.
5. Write a program to validate C identifiers and keywords.
6. Write a program to implement NFA such that it accepts strings that either end with 01 or end with 10, over alphabet $\Sigma=\{0,1\}$. Your program should ask the user to input the string and check whether the user input is accepted or rejected by the NFA. Also show the processing of the string in the output section.
7. Write a program to implement an NFA such that it accepts strings that start with 1 and end with 0, over alphabet $\Sigma=\{0,1\}$. Your program should ask the user to input the string and check whether the user input is accepted or rejected by the NFA. Also show the processing of the string in the output section.
8. Write a program to implement an ϵ -NFA that accepts strings of the language $L = \{0^n 1^m ; n, m \geq 0\}$, over alphabet $\Sigma=\{0,1\}$. Your program should ask the user to input the string and check whether the user input is accepted or rejected by the ϵ -NFA. Also show the processing of the string in the output section.

Note: First draw the respective transition diagram, transition table, and the code section should consist of the state as mentioned in the diagram and table. For reference use KEC book lab section. Write code in your preferable programming language without use of importing external libraries like in Python. Bring your code executed laptops for demonstration along with the hand written codes in A4 paper.